

Să se determine ⁼¹⁼ gradul de exactitate, restul folosind teorema lui Peano și formula iterată pentru

- regula MIDPOINT
- regula trapezului (trapezoidal rule)
- regula Simpson
- regula Simpson 3/8

Rezolvare

Ordinul unei formule de cuadratură

$$\int_a^b f(x) dx = \sum_{i=0}^n A_i f(x_i) + R(f)$$

A_i - coeficienți / ponderi

x_i - noduri

$R(f)$ - termenul rest / eroare

este " n " dacă este "exactă" pt. $1, x, \dots, x^n$, $n \in \mathbb{N}$
adică $\int_a^b x^p dx = \sum_{i=0}^n A_i x_i^p$, $p = \overline{0, n}$

$$(R(x^p) = 0, p = \overline{0, n})$$

$$\text{și } \int_a^b x^{n+1} dx \neq \sum_{i=0}^n A_i x_i^{n+1}$$

$$(R(x^{n+1}) \neq 0)$$

Formulele de cuadratură menționate se obțin folosind interpolarea Lagrange cu noduri echidistante din $[a, b]$

(I) $n=0 \Rightarrow$ avem un singur nod $x_0 \in (a, b]$

$$L_0(x_0; f)(x) \stackrel{\text{not}}{=} L_0(x) = f(x_0) = \text{constant}$$

(polinom de grad 0)

$$\int_a^b f(x) dx \approx \int_a^b L_0(x) dx = \int_a^b f(x_0) dx = (b-a) f(x_0)$$

$$\Rightarrow (1) \int_a^b f(x) dx \approx (b-a) f(x_0) = A_0 f(x_0) \text{ cu } A_0 = b-a$$

Observație !! $\Rightarrow (2) \int_a^b f(x) dx = (b-a) f(x_0) + R_d(f)$

Dacă formula de cuadratură are gradul de exactitate n atunci $R(P) = 0 \quad \forall P \in \Pi_n$ și există $Q \in \Pi_{n+1}$ a.i. $R(Q) \neq 0$

În (3) considerăm (2) se mai numește și regula dreptunghiului

$$f \equiv 1 \equiv x^0 \Rightarrow \int_a^b 1 dx = (b-a) \cdot 1 + R_d(1)$$

$$\Rightarrow R_d(1) = 0$$

$$f \equiv x^1 \Rightarrow \int_a^b x dx = (b-a) \cdot x_0 + R_d(x)$$

$$\Rightarrow (b-a) x_0 + R_d(x) = \frac{b^2 - a^2}{2}$$

$$\Rightarrow R_d(x) = (b-a) \left(\frac{a+b}{2} - x_0 \right)$$

$$\Rightarrow \left(R_d(x) = 0 \Leftrightarrow \frac{a+b}{2} = x_0 \right)$$

În concluzie formula (2) are gradul de exactitate $\begin{cases} 0, & \text{dacă } x_0 \neq \frac{a+b}{2} \\ 1, & \text{dacă } x_0 = \frac{a+b}{2} \end{cases}$

$$\Rightarrow (3) \int_a^b f(x) dx = (b-a) f\left(\frac{a+b}{2}\right) + R_d(f)$$

↑ MIDPOINT RULE

(are gradul de exactitate 1)

$$f \equiv x^2 \Rightarrow \int_a^b x^2 dx = (b-a) \left(\frac{a+b}{2} \right)^2 + R_d(x^2) \Rightarrow \frac{b^3 - a^3}{3} = \frac{(b-a)(a+b)^2}{4} + R_d(x^2)$$

$\Rightarrow R_d(x^2) \neq 0$

=3

(II) $n=1 \Rightarrow$ avem două noduri $x_0, x_1 \in [a, b]$

Considerăm $x_0=a, x_1=b$ și $h=b-a$

$$\Rightarrow L_1(x_0, x_1; f)(x) = L_1(a, b; f)(x) \stackrel{\text{not}}{=} L_1(x) =$$

$$= \frac{x-b}{a-b} f(a) + \frac{x-a}{b-a} f(b) \quad (\text{polinom de grad 1})$$

$$\int_a^b f(x) dx \approx \int_a^b L_1(x) dx = \int_a^b \left(\frac{x-b}{a-b} f(a) + \frac{x-a}{b-a} f(b) \right) dx =$$

$$= \frac{b-a}{2} (f(a) + f(b))$$

$$\Rightarrow (4) \quad \int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b)) = A_0 f(x_0) + A_1 f(x_1)$$

$$\text{cu } A_0 = A_1 = \frac{b-a}{2}, \quad x_0=a, x_1=b$$

$$(5) \quad \int_a^b f(x) dx = \frac{b-a}{2} f(a) + \frac{b-a}{2} f(b) + R_f(f)$$

↑ Regula trapezului

În (5) considerăm

$$f \equiv 1 \equiv x^0 \Rightarrow \int_a^b 1 dx = \frac{b-a}{2} \cdot 1 + \frac{b-a}{2} \cdot 1 + R_f(1)$$

$$\Rightarrow R_f(1) = 0$$

$$f \equiv x^1 \Rightarrow \int_a^b x dx = \frac{b-a}{2} \cdot a + \frac{b-a}{2} b + R_f(x)$$

$$\Rightarrow \frac{x^2}{2} \Big|_a^b = \frac{b-a}{2} (a+b) + R_f(x) \Rightarrow \frac{b^2-a^2}{2} = \frac{b^2-a^2}{2} + R_f(x)$$

$$\Rightarrow R_f(x) = 0$$

$$f \equiv x^2 \Rightarrow \int_a^b x^2 dx = \frac{b-a}{2} a^2 + \frac{b-a}{2} b^2 + R_f(x^2)$$

$$\Rightarrow \frac{x^3}{3} \Big|_a^b = \frac{b-a}{2} (a^2 + b^2) + R_f(x^2) \Rightarrow \frac{b^3-a^3}{3} = \frac{a^2b-a^3+b^3-ab^2}{2} +$$

$$+ R_f(x^2) \Rightarrow R_f(x^2) \neq 0$$

=4=

!!! Regula trapezului are gradul de exactitate 1 !!!

(II) $n=2 \Rightarrow$ avem trei noduri $x_0, x_1, x_2 \in [a, b]$

Considerăm $x_0 = a, x_1 = \frac{a+b}{2}, x_2 = b; h = \frac{b-a}{2}$

$$\Rightarrow L_2(x_0, x_1, x_2; f) = L_2(a, \frac{a+b}{2}, b; f)(x) \stackrel{\text{not.}}{=} L_2(x) =$$

$$= \frac{(x - \frac{a+b}{2})(x-b)}{(a - \frac{a+b}{2})(a-b)} f(a) + \frac{(x-a)(x-b)}{(\frac{a+b}{2} - a)(\frac{a+b}{2} - b)} f(\frac{a+b}{2}) + \frac{(x-a)(x - \frac{a+b}{2})}{(b-a)(b - \frac{a+b}{2})} f(b)$$

(polinom de gradul al doilea)

$$\int_a^b f(x) dx \approx \int_a^b L_2(x) dx = \int_a^b \left(\frac{(x - \frac{a+b}{2})(x-b)}{(a - \frac{a+b}{2})(a-b)} f(a) + \right.$$

$$\left. + \frac{(x-a)(x-b)}{(\frac{a+b}{2} - a)(\frac{a+b}{2} - b)} f(\frac{a+b}{2}) + \frac{(x-a)(x - \frac{a+b}{2})}{(b-a)(b - \frac{a+b}{2})} f(b) \right) dx =$$

$$= \dots = \frac{b-a}{6} (f(a) + 4f(\frac{a+b}{2}) + f(b))$$

$$\Rightarrow (6) \int_a^b f(x) dx \approx \frac{b-a}{6} (f(a) + 4f(\frac{a+b}{2}) + f(b)) \equiv A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2)$$

cu $A_0 = \frac{b-a}{6}, A_1 = \frac{4(b-a)}{6}, A_2 = \frac{b-a}{6}, x_0 = a, x_1 = \frac{a+b}{2}, x_2 = b$

$$(7) \int_a^b f(x) dx = \frac{b-a}{6} f(a) + \frac{2(b-a)}{3} f(\frac{a+b}{2}) + \frac{b-a}{6} f(b) + R_S(f)$$

↑ Formula lui Simpson

În (7) considerăm

$$f \equiv 1 \Rightarrow \int_a^b 1 dx = \frac{b-a}{6} \cdot 1 + \frac{2}{3}(b-a) \cdot 1 + \frac{b-a}{6} \cdot 1 + R_S(1)$$

$$\Rightarrow b-a = \frac{b-a}{6} (1+4+1) + R_S(1) \Rightarrow R_S(1) = 0$$

$$f \equiv x \Rightarrow \int_a^b x dx = \frac{b-a}{6} a + \frac{2}{3}(b-a) \frac{a+b}{2} + \frac{b-a}{6} b + R_S(x)$$

$$\frac{x^2}{2} \Big|_a^b = \frac{b-a}{6} (a + 4 \frac{a+b}{2} + b) + R_S(x)$$

$$\frac{b^2-a^2}{2} = \frac{b-a}{6} (3a+3b) + R_S(x) \Rightarrow R_S(x) = 0$$

$$f \equiv x^2 \Rightarrow \int_a^b x^2 dx = \frac{b-a}{6} \left(a^2 + \frac{1}{3}(b-a) \left(\frac{a+b}{2} \right)^2 + \frac{b-a}{6} b^2 \right) + R_5(x^2)$$

$$\Rightarrow \frac{x^3}{3} \Big|_a^b = \frac{b-a}{6} \left(a^2 + 4 \left(\frac{a+b}{2} \right)^2 + b^2 \right) + R_5(x^2)$$

$$\Rightarrow \frac{b^3-a^3}{3} = \frac{b-a}{6} (2a^2 + 2b^2 + 2ab) + R_5(x^2)$$

$$\Rightarrow \frac{b^3-a^3}{3} = \frac{(b-a)(a^2+ab+b^2)}{3} + R_5(x^2) \Rightarrow R_5(x^2) = 0$$

$$f \equiv x^3 \Rightarrow \int_a^b x^3 dx = \frac{b-a}{6} \left(a^3 + 4 \left(\frac{a+b}{2} \right)^3 + b^3 \right) + R_5(x^3)$$

$$\frac{x^4}{4} \Big|_a^b = \frac{b-a}{6} \left(a^3 + \frac{a^3 + 3a^2b + 3ab^2 + b^3}{2} + b^3 \right) + R_5(x^3)$$

$$\frac{b^4-a^4}{4} = \frac{b-a}{12} (3a^3 + 3a^2b + 3ab^2 + 3b^3) + R_5(x^3)$$

$$b^4-a^4 = (b-a)(a^3+a^2b+ab^2+b^3) + R_5(x^3) \Rightarrow R_5(x^3) = 0$$

$$f \equiv x^4 \Rightarrow \int_a^b x^4 dx = \frac{b-a}{6} \left(a^4 + 4 \left(\frac{a+b}{2} \right)^4 + b^4 \right) + R_5(x^4)$$

$$\frac{x^5}{5} \Big|_a^b = \frac{b-a}{6} \left(a^4 + \frac{a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4}{4} + b^4 \right) + R_5(x^4)$$

$$\frac{b^5-a^5}{5} = \frac{b-a}{24} (5a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 5b^4) + R_5(x^4)$$

$$\Rightarrow R_5(x^4) \neq 0$$

!!! Regula lui Simpson are gradul de exactitate 3!!!

(IV) $m=3$ avem 4 noduri $x_0, x_1, x_2, x_3 \in [a, b]$

Considerăm $x_0=a$, $x_1=\frac{2a+b}{3}$, $x_2=\frac{a+2b}{3}$, $x_3=b$, $h=\frac{b-a}{3}$

$$\Rightarrow L_3(x_0, x_1, x_2, x_3; f) = L_3\left(a, \frac{2a+b}{3}, \frac{a+2b}{3}, b; f\right)(x) = L_3(x) =$$

$$= \frac{(x-\frac{2a+b}{3})(x-\frac{a+2b}{3})(x-b)}{(a-\frac{2a+b}{3})(a-\frac{a+2b}{3})(a-b)} f(a) + \frac{(x-a)(x-\frac{a+2b}{3})(x-b)}{(\frac{2a+b}{3}-a)(\frac{2a+b}{3}-\frac{a+2b}{3})(\frac{2a+b}{3}-b)} f(\frac{2a+b}{3})$$

$$= 6 =$$

$$+ \frac{(x-a)(x-\frac{2a+b}{3})(x-b)}{(\frac{a+2b}{3}-a)(\frac{a+2b}{3}-\frac{2a+b}{3})(\frac{a+2b}{3}-b)} f(\frac{2a+b}{3}) +$$

$$+ \frac{(x-a)(x-\frac{a+2b}{3})(x-\frac{2a+b}{3})}{(b-a)(b-\frac{2a+b}{3})(b-\frac{a+2b}{3})} f(b) \quad (\text{polinom de gradul al treilea})$$

$$\int_a^b f(x) dx \approx \int_a^b L_3(x) dx = \dots =$$

$$= \frac{b-a}{8} \left(f(a) + 3f(\frac{2a+b}{3}) + 3f(\frac{a+2b}{3}) + f(b) \right)$$

$$\Rightarrow (8) \int_a^b f(x) dx \approx \frac{b-a}{8} \left(f(a) + 3f(\frac{2a+b}{3}) + 3f(\frac{a+2b}{3}) + f(b) \right) =$$

$$= A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + A_3 f(x_3)$$

$$\text{cu } A_0 = \frac{b-a}{8}, A_1 = \frac{3(b-a)}{8}, A_2 = \frac{3(b-a)}{8}, A_3 = \frac{b-a}{8}$$

$$x_0 = a, x_1 = \frac{2a+b}{3}, x_2 = \frac{a+2b}{3}, x_3 = b$$

$$(9) \int_a^b f(x) dx = \frac{b-a}{8} f(a) + \frac{3(b-a)}{8} f(\frac{2a+b}{3}) + \frac{3(b-a)}{8} f(\frac{a+2b}{3}) + \frac{b-a}{8} f(b) + R_{3/8}(f)$$

~~Formula~~ Formula lui Simpson 3/8

$$f \equiv 1 = x^0 \Rightarrow \int_a^b 1 dx = \frac{b-a}{8} \cdot 1 + \frac{3(b-a)}{8} \cdot 1 + \frac{3(b-a)}{8} \cdot 1 + \frac{b-a}{8} \cdot 1 + R_{3/8}(1)$$

$$\Rightarrow b-a = \frac{b-a}{8} (1+3+3+1) + R_{3/8}(1) \Rightarrow R_{3/8}(1) = 0$$

$$f \equiv x \Rightarrow \int_a^b x dx = \frac{b-a}{8} a + \frac{3(b-a)}{8} \frac{2a+b}{3} + \frac{3(b-a)}{8} \frac{a+2b}{3} +$$

$$+ \frac{b-a}{8} \cdot b + R_{3/8}(x) \Rightarrow \frac{b^2-a^2}{2} = \frac{b-a}{8} \left(a + 3 \frac{2a+b}{3} + 3 \frac{a+2b}{3} + b \right) + R_{3/8}(x)$$

$$\Rightarrow \frac{b^2-a^2}{2} = \frac{b-a}{8} (4a+4b) + R_{3/8}(x)$$

$$\Rightarrow \frac{b+a}{2} = \frac{a+b}{2} + R_{3/8}(x) \Rightarrow R_{3/8}(x) = 0$$

$$\begin{aligned}
 f &\equiv x^2 \Rightarrow \int_a^b x^2 dx = \frac{b-a}{8} \left(a^2 + 3 \left(\frac{2a+b}{3} \right)^2 + 3 \left(\frac{a+2b}{3} \right)^2 + b^2 \right) + R_{3/8}(x^2) \\
 \frac{b^3-a^3}{3} &= \frac{b-a}{8} \left(a^2 + \frac{4a^2+4ab+b^2}{3} + \frac{a^2+4ab+4b^2}{3} + b^2 \right) + R_{3/8}(x^2) \\
 \Rightarrow \frac{a^2+ab+b^2}{3} &= \frac{1}{8} (8a^2+8ab+8b^2) + R_{3/8}(x^2) \Rightarrow R_{3/8}(x^2) = 0
 \end{aligned}$$

$$\begin{aligned}
 f &\equiv x^3 \Rightarrow \int_a^b x^3 dx = \frac{b-a}{8} \left(a^3 + 3 \left(\frac{2a+b}{3} \right)^3 + 3 \left(\frac{a+2b}{3} \right)^3 + b^3 \right) + R_{3/8}(x^3) \\
 \frac{b^4-a^4}{4} &= \frac{b-a}{8} \left(a^3 + \frac{8a^3+12a^2b+6ab^2+b^3}{9} + \frac{a^3+12ab^2+6a^2b+4b^3}{9} + b^3 \right) + R_{3/8}(x^3)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow b^3+b^2a+ba^2+a^3 &= \frac{1}{2} (2a^3+2a^2b+2ab^2+2b^3) + R_{3/8}(x^3) \\
 \Rightarrow R_{3/8}(x^3) &= 0
 \end{aligned}$$

$$\begin{aligned}
 f &\equiv x^4 \Rightarrow \int_a^b x^4 dx = \frac{b-a}{8} \left(a^4 + 3 \left(\frac{2a+b}{3} \right)^4 + 3 \left(\frac{a+2b}{3} \right)^4 + b^4 \right) + R_{3/8}(x^4) \\
 \frac{b^5-a^5}{5} &= \frac{b-a}{8} \left(a^4 + \frac{16a^4+32a^3b+24a^2b^2+8ab^3+b^4}{27} + \frac{a^4+8a^3b+24a^2b^2+32ab^3+b^4}{27} + b^4 \right) + R_{3/8}(x^4) \\
 \Rightarrow \frac{b^5-a^5}{5} &= \frac{b-a}{8} \left(\frac{4a^4+40a^3b+48a^2b^2+40ab^3+44b^4}{27} \right) + R_{3/8}(x^4) \\
 \Rightarrow R_{3/8}(x^4) &\neq 0
 \end{aligned}$$

!!! Regula Simpson 3/8 are gradul de exactitate 3!!!

CONCLUZIE PT GRADUL DE EXACTITATE

- dreptunghiului: 0

- trapez: 1

- Simpson 3/8: 3

- Midpoint: 1

- Simpson: 3